

Delhivery Feature Engineering Project Report

■ Project Overview

This project focuses on data cleaning, feature engineering, and exploratory analysis of Delhivery's logistics dataset. The objective is to transform raw operational data into structured analytical features that can help optimize routes and improve delivery efficiency.

■ Objective

- Process and clean raw logistics data.
- Engineer new time and location-based features.
- Perform exploratory data analysis (EDA) to find delivery trends.
- Conduct hypothesis testing to validate operational assumptions.
- Generate actionable business insights for optimization.

■ Process Summary

1. **Data Cleaning:** Removed irrelevant columns, handled missing values, converted timestamps.
2. **Feature Engineering:** Created features such as total trip time, day, month, week, hour, and parsed source/destination locations.
3. **EDA:** Analyzed route types, time patterns, and city/state-wise distribution of trips.
4. **Hypothesis Testing:** Used t-tests to compare estimated (OSRM) vs. actual times and distances.
5. **Scaling:** Normalized and standardized numeric features for modeling readiness.

■ Key Insights

- Majority of trips were through **Carting** (60%) compared to FTL (40%).
- Peak delivery creation time: **6 PM – 10 PM**.
- Top source cities: Mumbai, Gurgaon, Delhi, Bengaluru.
- Top destination cities: Mumbai, Bengaluru, Gurgaon, Delhi.
- States with highest activity: Maharashtra, Karnataka, Haryana, Tamil Nadu.
- Statistical testing revealed that OSRM-estimated times differ significantly from actual delivery times.

■ Business Recommendations

- Optimize the OSRM route estimation system to reduce delivery-time discrepancies.
- Investigate route deviations and enhance traffic-based prediction accuracy.
- Strengthen delivery corridors in high-demand cities and states.
- Profile customers and hubs in Maharashtra, Karnataka, and Haryana for deeper insights.
- Use engineered features to build predictive models for delivery optimization.

■ Conclusion

The project successfully transformed Delhivery's logistics data into a clean, analytical dataset of 14,817 trips with 29 engineered features. Insights from the analysis can help improve operational efficiency and enhance customer experience through data-driven decisions.

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